DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING(A)
Osmania University, Hyderabad-07

CLASS TEST-II

SUBJECT: Data Structures

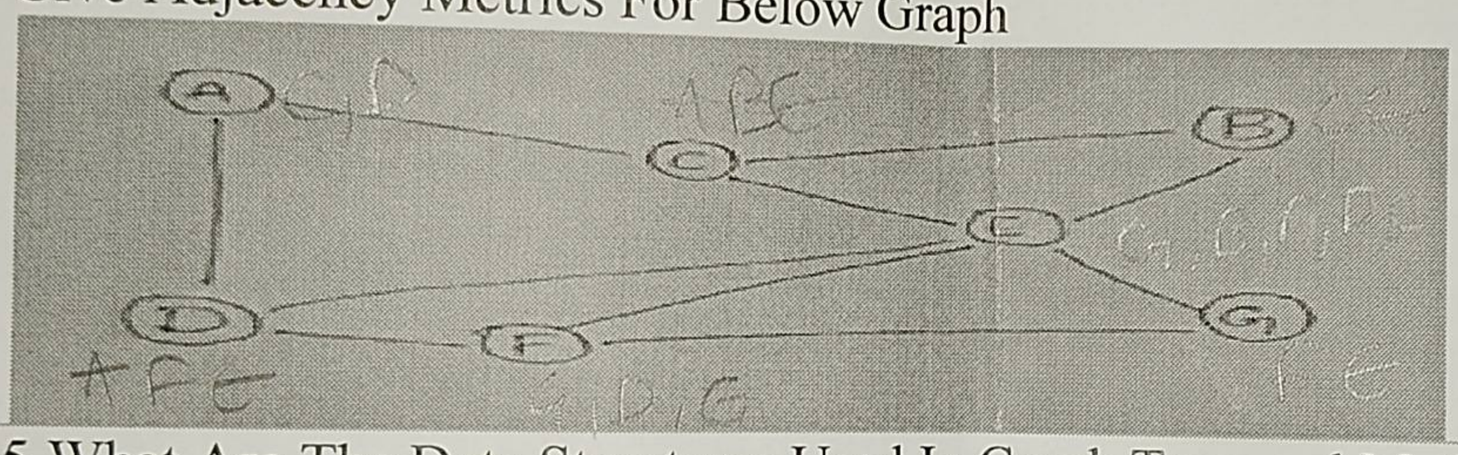
Class: B.E (CSE/AIML)

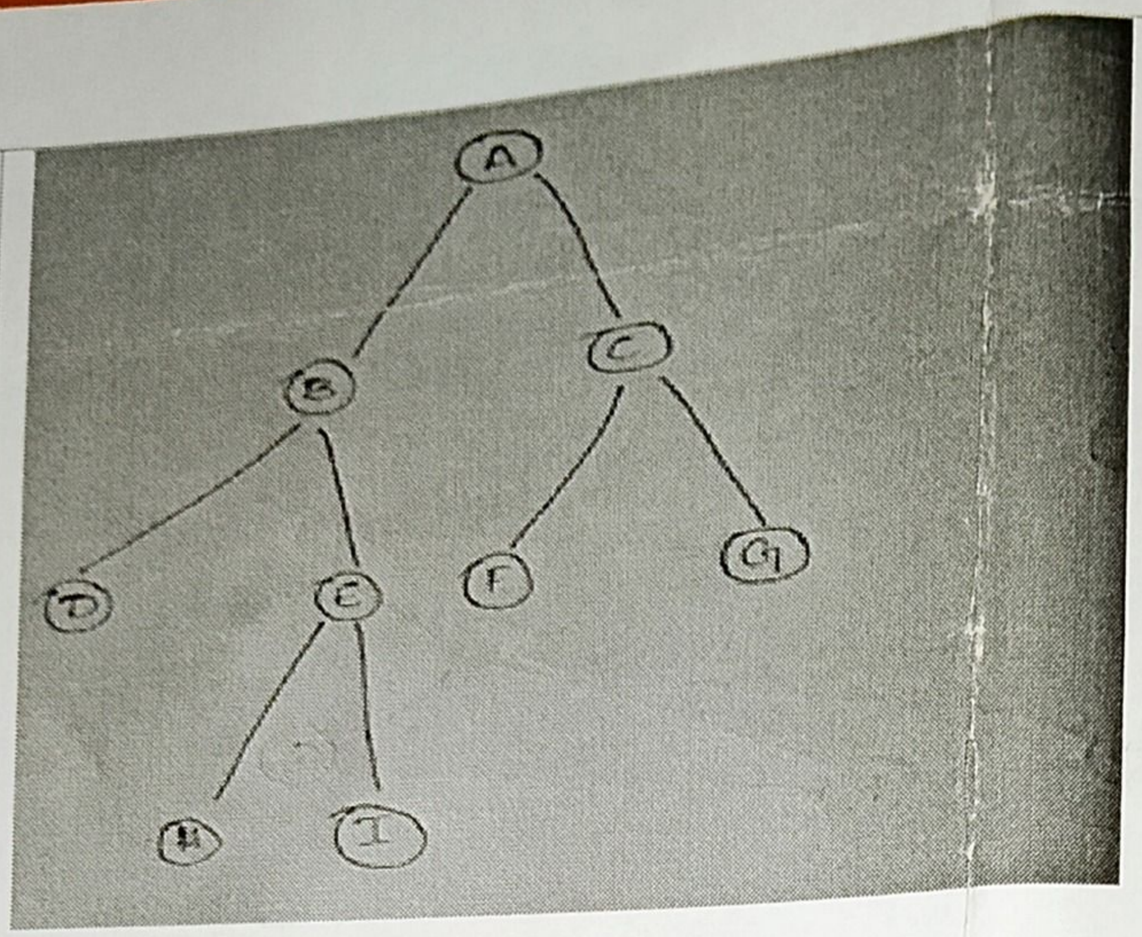
Semester: III

Date :25-11-2025

Max Marks: 30

Duration: 1.5 Hour

		Marks	BT	CO
	PART A **Answer All Questions (10 M)**			
1. a)	Mention All The Applications Of Stacks	2	3	3
b)	State The Properties Of Binary Tree	2	2	3
c)	Why Do We Need AVL Tree & Define Balance Factors	2	1	4
d)	Give Adjacency Metrics For Below Graph 	2	2	4
e)	5.What Are The Data Structures Used In Graph Traversal Methods	2	2	2
	PART B **Answer Any Two Questions (20 M)**			
1.	a) Illustrate The QUEUE Operations Using Linked List b] How Infix To Postfix Conversion Is Done ? Show With An Example	5 5	2 3	3 3
2.	a] Write Kruskal's Algorithm And Give Example To Find MST ? b) Show In order , Preorder, Post order, Traversals For Below Figure	5 5	3 2	4 4



3.	Construct B Tree Of Order – 5 ? a] 2,7,9,23,45,1,5,14,25,13,8,19,31,56 [Show Step/Step] ? b] Delete 5,7,14 ?	10	3	2